

Numerical Simulations of Special Solutions to the Three-Body Problem

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ABSTRACT

The three-body problem examines the motion of three masses interacting through gravity. Unlike the two-body problem, which has closed-form solutions, the three-body problem has no general closed form solution. This is because the three-body problem is a chaotic system which makes it unpredictable, but it is possible to use numerical methods to get an estimate. I utilized widely known Newtonian mechanics to compute the forces and acceleration of the bodies and integrated using the leapfrog integration technique to ensure the simulation follows energy conservation. Exceptions have been found for the three-body problem these are called “special solutions” they have known analytical solutions. These solutions constrain the initial conditions for the three-body system by implementing specific mass ratios, positions, and velocities, causing unique periodic trajectories that can be numerically solved.

OBJECTIVES

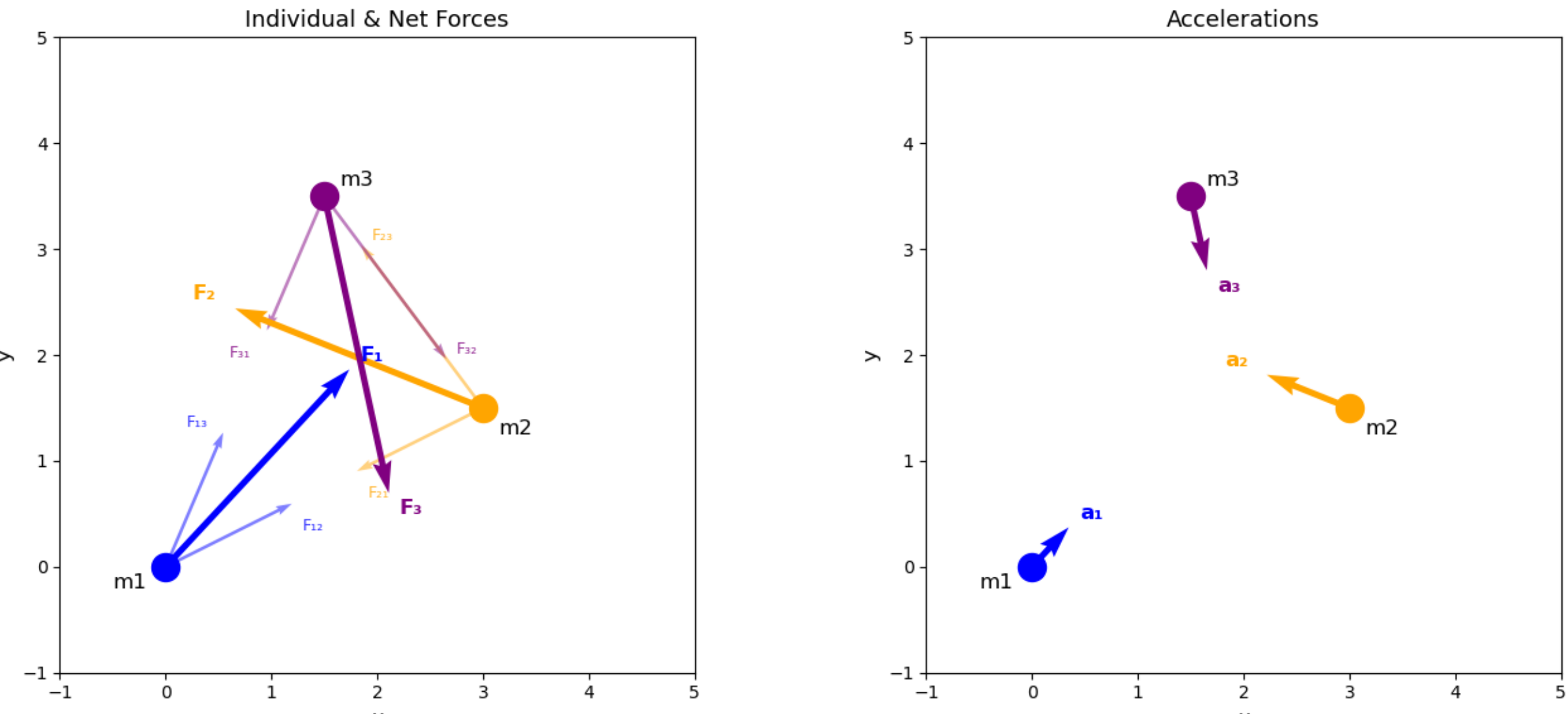


Figure 1:
Left: Shows the gravitational force vectors acting on each body by the other two bodies and the net gravitational force vector.
Right: Accelerations of each body caused by the gravitational force..

The three-body problem originates from the work of Isaac Newton in the 17th century. After successfully describing two-body motion, Newton attempted to extend his methods to systems of three mutually interacting bodies. He found that the problem becomes significantly more complex, with no general closed-form solution. Figure 1 illustrates this showing the force of the bodies depend on the net gravitational force, which is determined by the relative position of the other two bodies. Since the position of the bodies is always changing the forces continuously update meaning the equations are coupled. Since the equations are coupled and the force is non-linear(Equation.2), renders it difficult to find a general analytical solution. This is where special solutions take place these restrict or constrain the initial conditions to where the three-body problem is solvable. The objective of this work is to simulate several of the known special solutions to the three-body problem.

METHODS

1) $T^2 = \frac{4\pi^2}{G(M+m)}a^3$

2) $\vec{F}_{12} = -G \frac{m_1 m_2}{r^3} \vec{r}$

3) $\mathbf{a}_i = \sum_{j \neq i} -Gm_j \frac{\mathbf{r}_i - \mathbf{r}_j}{\|\mathbf{r}_i - \mathbf{r}_j\|^3}$

4) $E = \sum_i \frac{1}{2} m_i \|\mathbf{v}_i\|^2 - \sum_{i < j} \frac{Gm_i m_j}{\|\mathbf{r}_i - \mathbf{r}_j\|}$

Leapfrog Integration Technique

5) $\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i(t) + \mathbf{v}_i(t)\Delta t + \frac{1}{2} \mathbf{a}_i(t)\Delta t^2$

6) $\mathbf{v}_i(t + \Delta t) = \mathbf{v}_i(t) + \frac{1}{2} [\mathbf{a}_i(t) + \mathbf{a}_i(t + \Delta t)] \Delta t$

Variables:

T : Period
 a : Semi-Major Axis
 G : $6.67430 \times 10^{-11} m^3 kg^{-1} s^{-2}$
 M : Central Mass
 m : Outer Mass
 F : Force
 m_i, m_j : Masses
 r_i, r_j : Position Vectors
 E : Energy
 t : Time
 V_i : Velocity
 a_i : Acceleration
 Δt : Timestep = 0.001 seconds

INITIAL CONDITIONS

The following initial conditions are for simplicity there is other conditions that may yield the same results

Collinear Solution

Initial Conditions	Body 1	Body 2	Body 3
Mass (Solar Mass)	1.0	1.0	1.0
Position (AU)	(-2.0, 0.0)	(0.0,0.0)	(2.0, 0.0)
Velocity (AU/Year)	(0.0, -4.967)	(0.0,0.0)	(0.0, 4.967)

Triangular Solution

Initial Conditions	Body 1	Body 2	Body 3
Mass (Solar Mass)	100.0	5.0	1.0
Position(AU)	(0.0377,0.0980)	(0.962, -1.634)	(1.037, -1.634)
Velocity (AU/Year)	(-2.24,0.86)	(37.38,-22.01)	(37.38,23.74)

Checiner-Montgomery Solution

Initial Conditions	Body 1	Body 2	Body 3
Mass (Solar Mass)	1.0	1.0	1.0
Position (AU)	(-0.970, 0.243)	(0.0,0.0)	(0.970, -0.243)
Velocity (AU/Year)	(2.929, 2.7165)	(-5.858, -5.433)	(2.929, 2.7165)

Sitnikov Problem

Initial Conditions	Body 1	Body 2	Body 3
Mass (Solar Mass)	1.0	1.0	0.0
Position (AU)	(1, 0, 0)	(-1, 0, 0)	(0, 0, 0.5)
Velocity (AU/Year)	(0,6.283,0)	(0,-6.283,0)	(0,0,0)

Butterfly Orbit

Initial Conditions	Body 1	Body 2	Body 3
Mass (Solar Mass)	1.0	1.0	1.0
Position (AU)	(-0.99902, 0.0)	(0.99902, 0.0)	(0.0, 0.0)
Velocity (AU/Year)	(0.347111, 0.532728)	(0.347111, 0.532728)	(-0.0694222, -1.065456)

Yin-Yang Orbit

Initial Conditions	Body 1	Body 2	Body 3
Mass (Solar Mass)	1.0	1.0	1.0
Position (AU)	(-1.0, 0.0)	(1.0, 0.0)	(0.0, 0.0)
Velocity (AU/Year)	(0.46444, 0.39606)	(0.46444, 0.39606)	(-0.92888, -0.79212)

Sources : Montgomery, R. (2019), Chenciner & Montgomery (2000), Misiak (n.d.)

Results

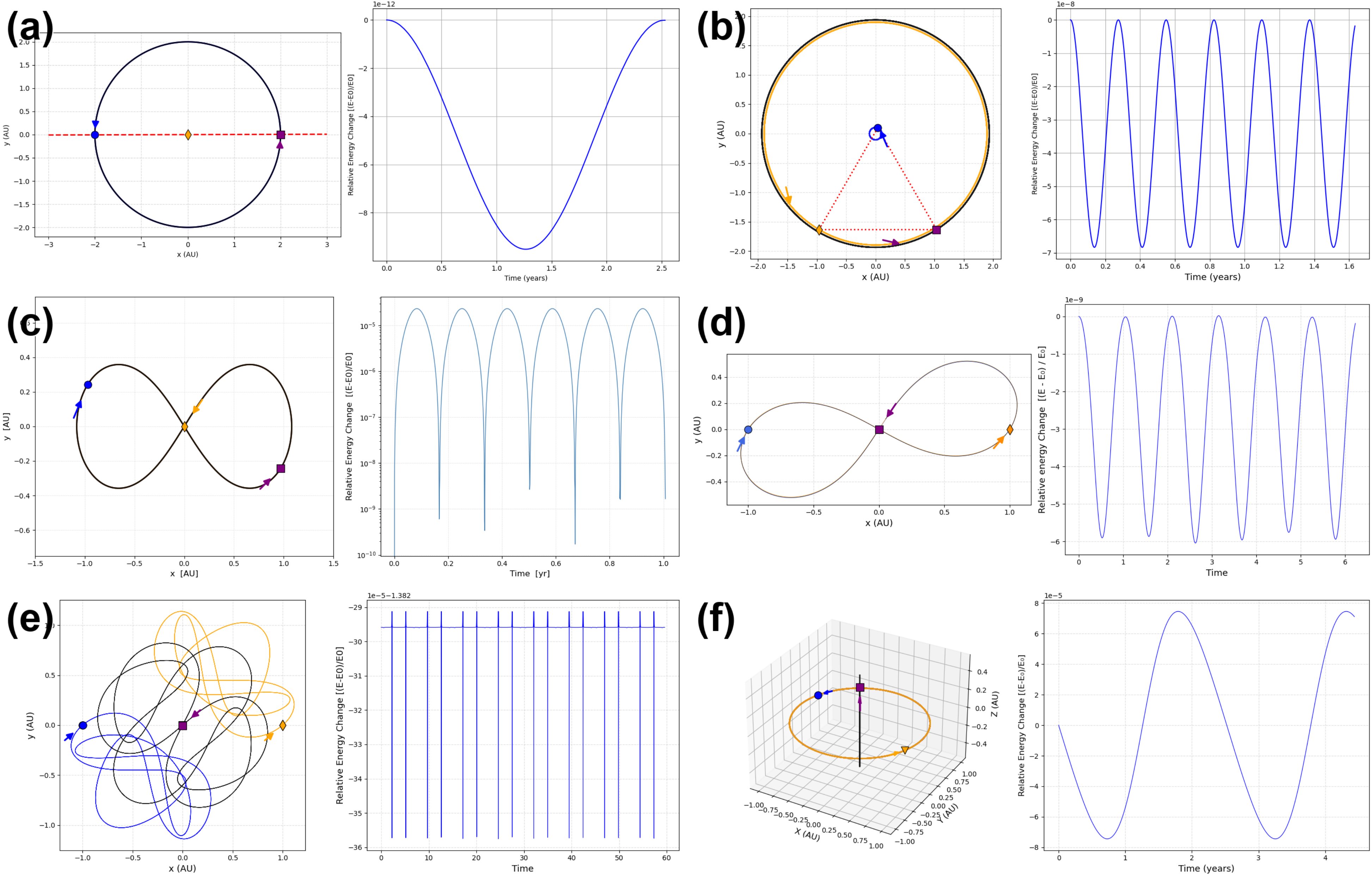


Figure 2: Trajectories and Relative Energy Change for Six Special Solutions;
(a) Collinear Solution, (b) Triangular Solution, (c) Checiner-Montgomery Solution, (d) Butterfly Orbit, (e) Yin-Yang Orbit, (f) Sitnikov Problem

CONCLUSIONS

Collinear

- The three bodies form a collinear relationship in their orbit confirming Euler’s prediction.
- The central mass remains Stationary while the outer bodies rotate symmetrically around it.

Triangular

- The Lagrange Solution yields a triangular relationship between the bodies
- The triangular starting positions keep their form throughout the simulation
- The outer bodies follow similar trajectories just with slightly different orbital radii

Chenciner-Montgomery Solution

- This solution is symmetric, so the center of mass naturally is at the origin
- Bodies follow the same exact path
- Lemniscate Orbit

Sitnikov Problem

- Massless third body oscillates along the z-axis
- The eccentricity must be equal to zero for periodic motion

Butterfly

- Resembles the Checiner-Montgomery solution just different orientation
- Bodies Pass through the Center twice per period

Yin-Yang

- Each body follows same unique path just phase shifted from one another
- Resembles Yin-Yang symbol

References

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3-D Simulations:



GitHub Repository:

